

Aniket Jain

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Summary

Data scientist with 4 years of data engineering and BI experience and a Master of Data Science. Hands-on with Python, pandas/NumPy, SQL (including PostgreSQL/Redshift), classical ML (regression, time series, model validation) and LLM/Generative AI prototyping and evaluation. Experienced building ML-ready pipelines, optimizing query and reporting performance, and collaborating with product, engineering and data teams to translate models and analytics into user-facing features.

Work Experience

Senior Software Developer

Sep 2019 - Aug 2023

Wipro Technologies

- Built ML-ready data pipelines and reusable preprocessing components in Python that combined validation logic and feature engineering to support modelling and analytical use cases.
- Evaluated machine learning and Generative AI approaches for business applications, documenting technical feasibility, limitations, testing results and production handover requirements to engineering teams.
- Designed and delivered an analytics-driven self-service Power BI product (daily refresh, incremental loads) that shifted recurring manual reporting to a reusable workflow; tracked reporting efficiency and improved report generation speed by approximately 20%, while reducing recurring manual effort by >10 hours/month.
- Optimized T-SQL queries and database structures for Microsoft SQL Server: used execution plans, query rewrites and indexing to reduce a representative reporting/data-processing query from ~40-60s to ~10-15s.
- Architected and implemented ETL pipelines and cloud migration transforms (on-prem → cloud), applying data modelling and performance tuning to deliver analysis-ready datasets for reporting and ML.
- Created training/validation/test datasets, applied cross-validation and standard performance metrics (precision, recall, F1, RMSE), and performed data quality checks to identify missing values, anomalies and duplicates.
- Implemented containerised analytics applications (Docker) and deployed to Kubernetes sandbox environments to improve reproducibility and safe pre-production testing.
- Partnered with product owners, data governance and engineering squads to translate business requirements into technical acceptance criteria and deployable solutions; led Agile ceremonies as a certified Scrum practitioner.

Projects

Stock Market Prediction Using ML & Econometrics

Personal / Academic Project

- Tested volatility modelling: ran ARCH-LM on 2,514 S&P 500 return observations ($LM = 975.79$, $p < 0.0001$) and selected GARCH(1,1) via AIC; reported coefficient estimates ($\alpha_1 = 0.1705$, $\beta_1 = 0.8264$) and validated forecasts against 30-day realised volatility with residual diagnostics.
- Compared ARIMA baseline vs multivariate LSTM (120-day sliding windows, technical indicators) using time-preserving train/test splits; ARIMA RMSE = 16.26 / MAE = 12.55, LSTM RMSE = 12.83 / MAE = 8.37 ($\approx 21\%$ RMSE reduction, $\approx 33\%$ MAE reduction); used results to recommend LSTM for point forecasts and explore hybrid LSTM-GARCH approaches.
- Applied standard time-series validation techniques (ADF, Ljung-Box), model selection via AIC, and monitored overfitting with dropout and early stopping.

Multi Agent Generative AI Workflow

Personal Project / Prototype

- Designed multi-agent workflows, tool-calling logic, prompt templates and output structure for research collection, summarisation and structured decision support using LLM frameworks.

- Developed an evaluation approach combining automated checks (workflow completion, correct tool usage, output format) with manual assessment of relevance, completeness and factual consistency; used iterative failure analysis to refine prompts, agent responsibilities and tool routing.
- Performed prototype-level qualitative annotation and pass/fail checks personally (no external annotators); used evaluation outcomes to stabilise workflows rather than issuing a formal production release metric.

Event-driven Data Integration with Apache Kafka

Personal Project

- Built an event-driven pipeline using Kafka with Python producers/consumers at tens of events/sec; partitioned topics by event type and used a business identifier as partition key to preserve ordering per entity.
- Implemented idempotent consumers (unique event_id checks, commit offsets only after successful processing) to allow safe replay and applied JSON payload patterns with schema-evolution guidance for production adoption.

E-commerce Performance Dashboard

Personal Project

- Cleaned and analysed orders, customers and product data; built an interactive Power BI dashboard surfaced with KPIs for revenue, order volume, customer trends and product performance to support business decisions.

Education

Master of Data Science and Decisions (Specialization: Business Data Science & Decisions)	Sep 2023 - Aug 2025
University of New South Wales	Sydney, Australia

Bachelor of Engineering in Computer Science & Engineering	Aug 2015 - Jun 2019
IES IPS Academy Indore	Indore

Certifications

Scrum Master
Power BI
AWS Certified AI Practitioner

Skills

Python • pandas • NumPy • SQL • PostgreSQL • AWS Redshift • Machine Learning • Time Series Forecasting • LSTM • GARCH • Experimental Design & Statistical Testing • Model Validation • MLflow • Docker • Kubernetes • AWS SageMaker • AWS Bedrock • LangChain • LlamaIndex • Prompt Engineering • Retrieval-Augmented Generation (RAG) • Power BI • Plotly • Matplotlib • Git • GitLab CI/CD • Apache Kafka • Ansible • Agile / Scrum

Languages

English • Hindi